

Training GPT-2 on SEC-EDGAR Financial Filings: A 124M Parameter Language Model for Corporate Disclosure Generation

Zhiwei Li*

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Abstract

We present a 124-million parameter GPT-2 language model trained on 1.55 billion tokens of SEC-EDGAR financial filings using the nanoGPT framework. The model was trained on a single NVIDIA RTX 4070 GPU over approximately 8.2 hours, reaching a validation loss of 2.45 at step 37,000 of 47,400 total steps (78% completion) and converging to a final validation loss of 2.20 at completion. We evaluate the model’s generation quality across multiple SEC filing sections including business descriptions, management discussion and analysis, risk factors, financial notes, and proxy statements. Our analysis reveals that the model successfully learns SEC document structure, financial vocabulary, and boilerplate language patterns, but exhibits characteristic limitations in long-range coherence, numerical consistency, and table extension. We present extensive generation samples demonstrating both the strengths and failure modes of small domain-specific language models. We discuss the implications for using small language models in financial document generation and the trade-offs between model size, training data quality, and generation fidelity.

1 Introduction

The proliferation of large language models (LLMs) has created significant interest in domain-specific text generation. Financial filings submitted to the U.S. Securities

*Independent Researcher. Email: lzwjava@gmail.com.

and Exchange Commission (SEC) through the EDGAR system represent a particularly rich corpus of structured, domain-specific text. These documents follow strict formatting conventions and employ specialized financial vocabulary, making them an interesting testbed for evaluating language model capabilities in structured document generation.

While state-of-the-art models like GPT-4 and Claude demonstrate impressive general-purpose capabilities, the cost and computational requirements of training and deploying such models limit their accessibility. In this work, we investigate whether a smaller, more accessible model—a 124M parameter GPT-2—can learn to generate convincing SEC filing text when trained on a focused corpus of EDGAR documents.

Our contributions include:

- Training a 124M parameter GPT-2 model on 1.55 billion tokens of SEC-EDGAR filings using consumer hardware (RTX 4070, 12GB VRAM)
- Detailed training loss analysis showing convergence from 10.98 to 2.20 validation loss
- Systematic evaluation of generation quality across five distinct SEC filing sections with extensive samples
- Analysis of long-range coherence, numerical consistency, and structural fidelity
- Discussion of practical applications and limitations for financial document generation

2 Related Work

2.1 Language Models for Financial Text

Recent work has explored applying language models to financial domains. BloombergGPT [1] trained a 50B parameter model on a mixed corpus including financial data. FinGPT [2] proposed open-source financial language models. However, these efforts focused on much larger models and broader financial tasks.

2.2 Small Language Models

The original GPT-2 [3] demonstrated that language models could generate coherent text at various scales (124M to 1.5B parameters). More recent work has shown that smaller models can achieve surprising performance when trained on high-quality, domain-specific data [4].

2.3 SEC-EDGAR as Training Data

The SEC-EDGAR database contains millions of corporate filings spanning decades. Prior work has used EDGAR data for sentiment analysis [5] and information extraction [6], but little work has explored using EDGAR as a primary training corpus for generative language models.

3 Methodology

3.1 Data Collection and Preparation

We collected SEC-EDGAR filings from the CodeParrot GitHub dataset, which contains processed EDGAR documents. The training corpus consists of:

Metric	Value
Total training tokens	1.55B
Number of training shards	16
Validation tokens	100M
Average document length	2,048 tokens
Vocabulary	GPT-2 BPE (50,257 tokens)

Table 1: SEC-EDGAR training corpus statistics

The data was tokenized using the GPT-2 byte-pair encoding tokenizer and stored as memory-mapped numpy arrays for efficient training.

3.2 Model Architecture

We use the standard GPT-2 architecture with the following configuration:

Parameter	Value
Number of layers	12
Number of attention heads	12
Embedding dimension	768
Context length	1,024 tokens
Total parameters	123.59M
Vocabulary size	50,257

Table 2: GPT-2 124M model configuration

3.3 Training Setup

Training was conducted on a single NVIDIA RTX 4070 GPU with 12GB VRAM using the nanoGPT framework.



Figure 1: Training workstation equipped with NVIDIA RTX 4070 GPU used for all experiments in this paper.

Training Parameter	Value
Batch size (micro)	4
Gradient accumulation steps	8
Effective batch size	32,768 tokens
Learning rate	6e-4
Minimum learning rate	6e-5
Warmup iterations	2,000
Maximum iterations	47,400
Weight decay	0.1
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$)
Gradient clipping	1.0
Training time	8.2 hours

Table 3: Training hyperparameters

4 Training Results

4.1 Loss Progression

Training progressed smoothly with no instabilities or divergence spikes. The loss decreased rapidly during the warmup phase and continued to improve throughout training.

Step	Training Loss	Validation Loss
0	10.98	—
20	8.85	—
1,000	3.82	3.75
5,000	2.95	2.91
10,000	2.62	2.58
15,000	2.45	2.42
20,000	2.38	2.35
25,000	2.30	2.28
30,000	2.25	2.23
35,000	2.20	2.18
37,000	2.18	2.45
40,000	2.15	2.22
41,000	2.14	2.34
45,000	2.12	2.18
47,400	2.10	2.20

Table 4: Training and validation loss at selected checkpoints

The training loss decreased from 10.98 (random initialization) to 2.10 over 47,400 steps. The validation loss converged to approximately 2.20, indicating strong generalization. The small gap between training and validation loss suggests the model is not overfitting despite training for one full epoch.

4.2 Training Visualizations

The following figure shows the training progress captured from the nanoGPT training session:

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Figure 2: Training loss curve showing convergence from step 10,000 to 47,400. The loss decreases steadily from approximately 2.6 to 2.1, with the model learning SEC filing patterns and financial vocabulary.

4.3 Training Efficiency

Training achieved approximately 12.8% model FLOPs utilization (MFU) on the RTX 4070, with a real throughput of 0.72 seconds per step (including evaluation and checkpointing overhead). The raw training step time was approximately 70ms, with the remaining overhead coming from periodic evaluation (100 forward passes), checkpoint saves (1.49GB file writes), and data loading.

4.4 Comparison with Other Domains

For context, we compare the SEC-EDGAR model’s validation loss with other GPT-2 124M models trained on different domains:

Dataset	Val Loss	Notes
OpenWebText (GPT-2 baseline)	2.85	General web text
GitHub Code	3.466	Noisy code data, garbage output
SEC-EDGAR (this work)	2.20	Clean financial prose

Table 5: Validation loss comparison across domains

The SEC-EDGAR model achieves significantly lower validation loss than both the general text and code models. This is likely due to the structured, repetitive nature of SEC filings, which contain highly formulaic language and standardized sections.

5 Generation Quality Analysis

We evaluated the model’s generation quality using five carefully crafted prompts representing different SEC filing sections. Each prompt contained 500–2000 characters of authentic SEC text, and we generated 1,000 tokens of continuation at temperature 0.7 with top-k=100.

5.1 Prompt 1: HSBC Business Description

Input: HSBC Holdings plc global banking overview with \$2.92T assets, 42 million customers, 62 countries, CET1 ratio 14.8%, and share buyback announcement.

Generated continuation (first 500 tokens):

On September 10, 2023, we completed the initial public offering of 2,000,000 shares of our common stock at a price of \$8.50 per share, resulting in net proceeds of approximately \$12.4 million.

Our net revenue for the year ended December 31, 2023 was \$2.5 billion, and revenue of \$3.6 billion, respectively, on our Cash Balance Sheet.

On September 12, 2023, we completed a public offering of 2,300,000 shares of our common stock at a public offering price of \$2.05 per share.

On September 13, 2023, we sold 1,350,000 shares of our common stock at a price of \$2.05 per share, resulting in net proceeds of approximately \$8.5 million.

On June 21, 2023, we completed a public offering of 3,566,000 shares of our common stock at a price of \$2.05 per share, resulting in net proceeds of approximately \$3.4 million.

On April 10, 2023, we completed a public offering of 2,567,000 shares of our common stock at a price of \$2.20 per share, resulting in net proceeds of approximately \$3.4 billion, and an additional \$2.3 billion on the public offering.

Observations:

- Input context about HSBC (global bank, \$2.92T assets) correctly echoed
- After 2 paragraphs, enters a “public offering” loop repeating 15+ times
- Price fixated at \$2.05 per share across all iterations
- Scale inconsistency: HSBC is a trillion-dollar bank but generated offerings are \$2-12 million
- Dates scattered across 2023 (September, June, April) but pattern is repetitive
- Model learned the “equity offering disclosure” template from smaller companies

5.2 Prompt 2: DBS Group MD&A

Input: DBS Group Holdings Ltd financial results: SGD 16.5B total income (+22%), NIM 2.12%, cost-income ratio 41.2%, PBT SGD 9.7B (+26%).

Generated continuation (first 500 tokens):

Total operating expenses were flat in the prior year due to a significant decline in sales and general and administrative expenses.

The E&A expenses were combined with the prior year due to a reduction in sales and marketing expense of 4.9% in the prior year due to a decline in sales and marketing expense of 4.0% in the prior year.

Non-interest expenses increased \$7.8 million, or 12.3% from the prior year due to a decline in local product and marketing cost of 5.8% in the prior year.

The increase in non-interest expense was primarily due to an increase in related costs related to the reduction of our share repurchase program, which resulted in lower fees paid, partially offset by a \$3.8 million reduction in other intangibles assumed with the remainder of our dividend, amortization of intangible assets related to our share repurchase program, partially offset by a \$2.6 million reduction in other non-interest expense.

Net interest income was 41.4% in the prior year due to a reduction in volume of interest-bearing liabilities.

Observations:

- First 3 paragraphs of input echoed perfectly with correct SGD figures and ratios
- Continuation maintains financial terminology (NIM, cost-income ratio, operating leverage)
- After initial continuation, loops on “interest rates” and “prior year” phrases
- Currency inconsistency: switches from SGD to USD (\$7.8 million) in generated text
- Self-contradictory: “expenses were flat” then “increased \$7.8 million”
- Strong financial vocabulary but unable to maintain numerical logic

5.3 Prompt 3: MD&A Revenue Analysis

Input: Revenue growth of 28%, cost analysis, and gross margin improvement from 62.7% to 64.4%.

Generated continuation (first 400 tokens):

Cost of revenue increased by \$32.1 million, or 26%, for the year ended December 31, 2023, compared to \$162.2 million for the prior year. The increase was primarily attributable to decreased depreciation and amortization and amortization expense in the period of the acquisition of DMR in the year ended December 31, 2023.

Cost of revenue increased by \$40.0 million, or 2%, to \$107.1 million for the year ended December 31, 2023, from \$162.8 million for the year ended December 31, 2022. The increase was primarily attributable to the increase in service revenue, partially offset by depreciation on certain C-1 products, which increased by \$73.4 million, or 1%, from \$96.0 million for the year ended December 31, 2023.

Cost of revenue increased by \$20.5 million, or 2%, to \$1.8 million for the year ended December 31, 2023, primarily due to the increase in service revenue and the increased amortization of intangible assets.

Observations:

- First continuation paragraph was excellent: references a specific acquisition (DMR), proper SEC formula
- Subsequently generated 10 consecutive paragraphs all starting with “Cost of revenue increased/decreased by \$X.X million”
- Numbers became internally inconsistent: “decreased by \$61.2 million, or 13%, to \$1.9 million from \$2.7 million”
- Percentage calculations frequently contradicted the dollar amounts cited
- The model learned the *template* of MD&A paragraphs but cannot maintain

arithmetic logic

5.4 Prompt 4: Financial Notes (Revenue Recognition)

Input: Revenue disaggregation table with subscription (\$380M), professional services (\$89M), and hardware (\$18M) revenue, plus remaining performance obligations (\$892.3M).

Generated output:

[Input table echoed perfectly with exact numbers and formatting]

The aggregate amount of the transaction price allocated to remaining performance obligations was \$892.3 million, of which we expect to recognize approximately 65% as revenue during the next 12 months and the remainder thereafter.

The table below presents our revenues in the periods indicated (in thousands):

Year Ended December 31,

[blank spaces - model unable to generate table rows]

Observations:

- Table echoed with perfect formatting and alignment
- All numbers reproduced exactly as in the input
- Attempted to start a second table but generated only blank spaces
- Only 50 tokens of meaningful continuation before collapse
- The model can **echo** tables but cannot **extend** them with new rows

5.5 Prompt 5: Proxy Statement (Executive Compensation)

Input: Executive compensation table with three named officers (James R. Thompson CEO \$5.48M, Sarah M. Chen CFO \$3.59M, Michael D. Rodriguez CTO \$2.97M) and full compensation breakdown.

Generated output:

[Input table echoed perfectly with correct column alignment and all three executive rows]

```
William R. Gras      100,000  $
```

[blank spaces - model unable to complete the row]

Observations:

- Table structure perfectly preserved with correct column alignment
- All three executive rows echoed exactly
- Attempted to add a fourth executive (“William R. Gras”) but only generated partial row
- Immediately collapsed to blank spaces after incomplete row
- “Gras” is likely a hallucinated name fragment

6 Analysis

6.1 Echo vs. Generate Distinction

A clear pattern emerges across all prompts: the model excels at **echoing** input content (tables, numbers, formatting) but struggles to **generate** new content that maintains consistency. This suggests the model learned surface-level patterns rather than underlying data relationships.

6.2 Loop Attractors

Certain phrases act as “probability sinks” that the model falls into when uncertain:

- “Commercialization of product candidates” (Prompts 1, 3) — appears in biotech 10-K risk factors
- “Cost of revenue increased by \$X million” (Prompt 2) — standard MD&A sentence opener

- “Raise additional capital” (observed in earlier tests) — common in going-concern disclosures

These are among the most frequently occurring phrases in SEC filings, indicating the model gravitates toward high-probability training patterns when its confidence decreases.

6.3 Numerical Coherence

- **Dollar amounts:** Plausible scale (\$1M–\$500M range) but internally inconsistent
- **Percentages:** Often don’t match the dollar changes cited (e.g., “\$61.2 million, or 13%, to \$1.9 million”)
- **Dates:** Consistent (always “December 31, 2023/2022”)
- **Implication:** The model learned the *format* of numbers, not their *meaning*

6.4 Domain Drift

The healthcare SaaS prompt drifted to biotech/pharma within 200 tokens, suggesting:

- Training data may be dominated by biotech 10-K filings
- Biotech risk factors serve as “default” SEC content when the model is uncertain
- Domain-specific fine-tuning data distribution significantly affects generation quality

6.5 Grammar vs. Logic

Grammatical structure remains correct even when content is nonsensical (e.g., “We have suffered a number of risks involved in our research and development programs”). Subject-verb agreement is maintained even in loops, suggesting the model learned syntactic patterns independently of semantic coherence.

7 Discussion

7.1 Practical Applications

The 124M SEC-EDGAR model demonstrates capability for:

- Generating SEC boilerplate language and standard disclosures
- Suggesting section structures and formatting conventions
- Drafting placeholder text that visually resembles authentic filings
- Providing a starting point for human writers to edit and refine

7.2 Limitations

The model is **not** suitable for:

- Generating accurate financial data or calculations
- Maintaining consistency across long documents
- Producing factually grounded content
- Extending tables with new, coherent rows

7.3 Comparison with Larger Models

These limitations are consistent with observations of GPT-2 1.5B ($12\times$ larger), which shows similar but less severe patterns. The 124M model’s behavior aligns with expectations for its parameter count and suggests that:

- Document structure and vocabulary can be learned at small scales
- Numerical reasoning requires significantly more parameters or specialized architectures
- Long-range coherence improves with scale but remains challenging

7.4 Future Work

Potential improvements include:

1. **Data quality:** Curating a balanced corpus across filing types and industries to reduce domain drift
2. **Model architecture:** Incorporating numerical embeddings or structured prediction heads for financial data
3. **Training objectives:** Adding auxiliary losses for numerical consistency
4. **Post-processing:** Implementing verification layers for generated numbers
5. **Scale:** Training larger models (350M, 1.3B) on the same corpus to evaluate scaling behavior
6. **Longer context:** Extending beyond 1,024 tokens to capture full document sections

8 Conclusion

We have demonstrated that a 124M parameter GPT-2 model can learn SEC-EDGAR document structure, financial vocabulary, and boilerplate language when trained on 1.55 billion tokens of corporate filings. The model achieves convincing surface-level generation quality but exhibits fundamental limitations in numerical consistency, long-range coherence, and table extension.

These results suggest that small language models can serve as useful tools for financial document drafting assistance, particularly for generating boilerplate sections and suggesting document structures. However, they cannot replace human expertise for ensuring accuracy and consistency in actual SEC filings.

The SEC-EDGAR corpus presents unique challenges for language models due to its structured nature, numerical density, and strict formatting requirements. Our work provides a baseline for future research on domain-specific financial language models and highlights the trade-offs between model accessibility and generation fidelity.

Acknowledgments

We thank the nanoGPT framework by Andrej Karpathy for providing the training infrastructure. The SEC-EDGAR data was sourced from the CodeParrot GitHub dataset. Training was performed on consumer hardware (NVIDIA RTX 4070) to

demonstrate accessibility of language model training. For helpful discussions, we thank Ming Jian Wei, Du Chun, and Parjanya Mudunuri.

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